

Nicholas Mikulski

Professor Henderson

CSCI 325

9 November 2024

Ethical Dilemmas in the Field of Computer Science

When people discuss the topic of computer science, it is common to dive deep into the many ethical dilemmas and codes of ethics that come about. Computer science is one of the more important fields that should take ethical considerations more seriously than others, due to the nature of technology and the argument of how unreliable, reliable, or biased it can be. The digital age is now in full effect, and it can be both a positive and a negative factor for people. This paper will focus on some of the different dilemmas that can appear in the field of computer science, while also sharing ideas on how we can combat these problems and relate code of ethics biblically.

In my own opinion, I believe that the number one ethical dilemma in the computer science field is the question of what companies that we feed information to are doing with our information, and the problem with data theft. Within the computer science field, both of these dilemmas are a common problem that is shared between many different people and companies. The risk of having data being leaked or spread online is more common than the regular person thinks, and it is why I believe it is the biggest ethical dilemma. Ben Nancholas, a content specialist who has worked at well-known educational institutions, shares his thoughts about the issue of online data privacy, stating “On one hand, there is the pressing issue of vulnerabilities and rising cybercrime, including the increasing sophistication of hackers and the methods they

use to prey on users and the devastating consequences that identity theft and infrastructure breaches often cause.” (Nancholas paragraph 11). Nancholas shares that the rise of cyber theft and crime also contributes to the issue of data privacy. I think that this would personally affect me in the computer science workspace because if I were to be a designer in a tech company, I would have to take into consideration the different ways data breaches can break into my company systems. This would make me have to take different approaches in how I would want to implement systems to try and prevent data theft from happening. A Forbes article dives deeper into the issue of selling data, saying “Data is said to be the most valuable asset in the world, so protecting it is paramount. Today, user data is quickly becoming commoditized and sold against the user’s will” (Forbes paragraph 9). The Forbes article suggests that companies often value user’s data the most out of anything. It also states that the data we feed them is sold without our information to other data companies who would find it valuable. This is an issue I believe is important to discuss, especially in the work field I would like to work in because the information that I feed online could be spread and sold to other places where I would not want it to go, thus leading to an ethical dilemma. Expanding on the issue of theft, Matthew Sweeney, a contributing writer and author, shares his thoughts on the different types of hacking, stating “Ethical hacking usually refers to the cybersecurity practice of testing digital security systems for vulnerabilities to find out how to fortify them better against hackers. However, some would even argue that illegal hacking can be used ethically to expose corporate or government corruption” (Sweeney paragraph 12). Sweeney describes both sides of the argument of hacking in the workspace, saying it is both a good and bad thing depending on what exactly is happening. Ethical hacking is ethical and is something that I could do in the workspace to test the security measures of a

company's system. Though hacking into a company system with the intent to use the information gained to personally benefit the hacker is when the situation becomes an ethical dilemma.

How I propose that someone should prepare for these ethical dilemmas is to first put themselves in the situation of an unethical hacker and understand why someone would want to intentionally gain information from a system. I believe that it is important to think like the person who wants to intentionally cause harm and get information because if someone puts themselves in the shoes of a hacker, it will make them think of why the data breacher is harming. Another course of action someone can take is to implement system countermeasures with many different approaches. I feel that the more countermeasures are tested or implemented, the harder it will be for data to get stolen. Playing around with your system is also a good practice as it can usually lead to someone finding vulnerabilities within the system that they can then fix after their discovery. I feel that these dilemmas can cause serious issues in my future workspace, although the idea I proposed for preparing the ethical dilemmas is good to practice.

The ACM Code of Ethics and Professional Conducts lists many ethical principles that are important to understand and follow in the field of computer science. There is one that stands out to me the most, particularly "Avoid Harm". This relates to the ethical issues of hacking and why unethical hackers have the intent to cause harm by compromising unauthorized systems. This Code of Ethics is most definitely supported by Biblical principles. In the scripture of Luke 6:31, Jesus says, "Do to others as you would have them do to you", (NIV Luke 6:31). By treating others the way, you want to be treated, you are ensuring that only good is happening, as more than likely you do not want to inflict harm on yourself. This is a good biblical quote to relate to the issue of hacking as it makes one consider their choice of hacking unethically to then be hacked themselves. The IEEE Code of Ethics also states multiple good principles that I and

many others should follow in the computer science field. The principle that applied most to me is the second principle listed, “To treat all persons fairly and with respect, to not engage in harassment or discrimination, and to avoid injuring others.” Like the ACMD Code of Ethics, this principle discusses the ethical issue of intentionally causing harm towards others. In the scripture of Ephesians 4:32, Paul says to readers “Be kind and compassionate to one another, forgiving each other, just as in Christ God forgave you” (NIV Ephesians 4:32). Paul says there is no need to harm one another, and in the case that it happens, you shall forgive them in the way that Christ God has forgiven us. This biblical quote provided in Ephesians supports the ethical principle of treating everyone fairly and avoiding harm toward others because it states the same moral propositions as the IEEE principle.

The ethical dilemmas in the computer science field will undeniably keep growing as more people will enter the workspace. Keeping up with the Code of Ethics of computing is important, as it can lead to fewer conflicts in the future involving the workspace. By prioritizing good work ethics and values, we can come together in the workplace in a more friendly and trusting manner. As the digital age and technology advances, being able to have a strong ethical foundation can lead to everyone receiving benefits from each other for the greater good.

Works Cited

- “Council Post: 14 Tech-Related Ethical Concerns and How They Can Be Addressed.” *Forbes*, Forbes Magazine, 12 Aug. 2024,
www.forbes.com/councils/forbestechcouncil/2022/08/01/14-tech-related-ethical-concerns-and-how-they-can-be-addressed/.
- Nancholas, Ben. “Ethical Challenges in Computer Science.” *Keele University - Study Online*, 3 Apr. 2024, online.keele.ac.uk/ethical-challenges-in-computer-science/.
- Sweeney, Matthew. “Ethical Dilemmas in Computer Science.” *ZDNET*, 11 Feb. 2022, www.zdnet.com/education/computers-tech/ethical-dilemmas-computer-science/.